



Mars had deep-sea hydrothermal vents

Observations are being that the Martian planet could have harboured deep-sea hydrothermal vents in billions of years ago. <http://digit.in/MarsVents>



Air pollution turned birds black

Back in the 20th century, the air was so polluted over the US Rust Belt, they literally changed colour. <http://digit.in/BirdPill>

Score	Eyes	Verbal	Motor
1	Doesn't open eyes at all	Makes no sounds	No movement
2	Opens eyes to pain stimuli	Makes some noises	Extension / decerebrate posture
3	Opens eyes in response to sound	Incoherent words	Abnormal / decorticate posture
4	Opens eyes spontaneously	Confused or disoriented	Withdraws from pain
5	- NA -	Normal conversation	Able to push away from localised pain
6	- NA -	- NA -	Obeys commands to move

If they aren't, that's when you move on to testing level 3, and so on. It's the same for speech and movement – start at level 5 and 6, respectively, and then work your way down until the patient meets the criteria.

However, when it comes to measuring movement (which is usually the most affected by brain damage), we know we're going to have to do a little further explaining:

Basically, movement is broken up into six levels. Level six would score you six points on the GCS, and is how all of us score. Even if a patient is weak, and unable to move about normally, they might be able to follow a simple command, such as, "Grab my hand with your left hand". If they can do this, they're considered to be Level 6, despite being too weak to move anything else properly.

All the other levels are measured in response to pain. Level 5 is when a patient is not only able to move their body in response to pain, but also able to use their hands to push away the cause of the pain. Level 4 would be movement away, but an inability to use their limbs to push away the cause of the pain. Level 3 is when the body has a decorticate response to pain – this is when the arms come close to the chest with the elbows bent as shown in the adjoining image. Level 2 is when the response is to get into a decerebrate posture – shown in the image. The major difference from the decorticate posture is that the arms are straight

and the wrists turn outwards. Level 1, of course, is when there's no response whatsoever.

Thus, if a patient has perfect eye movement, is using incoherent words and is able to push away from localised pain (E4, V3, M5), they get a GCS of 12, which is a pretty good prognosis. Eight and under on the GCS is considered to be bad, and perhaps life threatening.

Unless the patient is in a medically induced coma, there usually is noth-

ing much that doctors can do, even today about someone who is comatose. When a patient slips into a coma, the first response of medical staff is to rush them to the intensive care unit (ICU), and to try and keep them alive. This usually involves monitoring the heartbeat, breathing, blood pressure and also brain activity. Even once the patient is stable, pretty much all doctors and nurses can do is to tend to physical needs. Tube feeding, maintaining the correct electrolyte balance, avoidance of bed sores, etc., is usually what is done. If the reason for the coma is known, such a failing vital organ, then attempts are made to fix the problem, but there's nothing much else we know about the coma state.

Can coma patients feel? Are they trapped inside their own heads, unable to communicate? We just don't know...

VEGETATIVE STATES

Vegetative states, or rather persistent vegetative states are usually seen in patients with severe brain damage. They're not the same as a coma, be-

